IBM DATA SCIENCE CAPSTONE PROJECT

Peer-Review Assessment Deck

SpaceX First Stage Landing Prediction

Predicting Falcon 9 First Stage Reusability & Launch Economics

End-to-end data science workflow analyzing orbital launch parameters, telemetry, interactive geospatial maps, and predictive classification models to determine landing success.

Author & Lead Data Scientist
Data Science Capstone Candidate

Project Deliverable Repository
github.com/spacex-capstone-report

PROJECT OVERVIEW

Executive Summary

Business Challenge

SpaceX advertises Falcon 9 launches at $62M vs. competitor costs of $165M+. First-stage reusability drives this $103M cost advantage. Predicting landing success enables commercial bidding optimization.

Methodology

Extracted REST API telemetry & Wikipedia tables. Performed SQL data wrangling, EDA, Folium interactive proximity mapping, Plotly Dash dashboarding, and trained 4 machine learning classifiers.

Key Outcome

Achieved 83.3% test accuracy across Logistic Regression, SVM, Decision Tree, and KNN models. Identified flight count, payload ranges (2k–5k kg), and launch site location as primary predictors.

83.3%
Best Model Accuracy Score
GridSearch tuned classifiers achieved equal top test set accuracy

$103M
Cost Delta per Launch
Savings unlocked by successful booster recovery

101
Total Launches Analyzed
Dataset spanning early testing through Block 5 operations

PROBLEM STATEMENT

Introduction & Business Value

SpaceX has revolutionized space logistics by making rocket launches reusable. The first stage booster accounts for the vast majority of total vehicle manufacturing cost.

1. **Disruptive Pricing Structure**
   Falcon 9 charges $62M per flight compared to legacy $165M+ commercial alternatives.

2. **Predictive Competitive Advantage**
   Determining booster landing probability allows commercial rivals to estimate exact launch margins.

3. **Strategic Bidding Models**
   Assists space ventures ("SpaceY") in placing optimal commercial bids against SpaceX payloads.



METHODOLOGY PHASE 1

Data Collection Pipeline

SpaceX REST API

Queried `api.spacexdata.com/v4/launches` endpoint using `requests.get()`.

- **Filtered Payload:** Isolated Falcon 9 rocket flights.
- **Nested Decoding:** Unpacked JSON objects for cores, payloads, rockets, and launchpads.
- **Custom Functions:** Extracted booster mass, orbit, and landing site.

Web Scraping (Wikipedia)

Scraped historical launch tables using `BeautifulSoup` HTML parsing.

- **Table Target:** Extracted Falcon 9 historical records.
- **Data Parsing:** Cleaned launch dates, payload masses, and mission outcomes.
- **Verification:** Cross-referenced API records for completeness.

Integrated Dataset

Constructed Pandas DataFrame exported to flat `.csv` format.

- **Flight Records:** 101 historical launch observations.
- **Feature Columns:** FlightNumber, Date, PayloadMass, Orbit, LaunchSite.
- **Ground Truth:** Raw outcome telemetry labels.

METHODOLOGY PHASE 2

Data Wrangling & Class Labeling

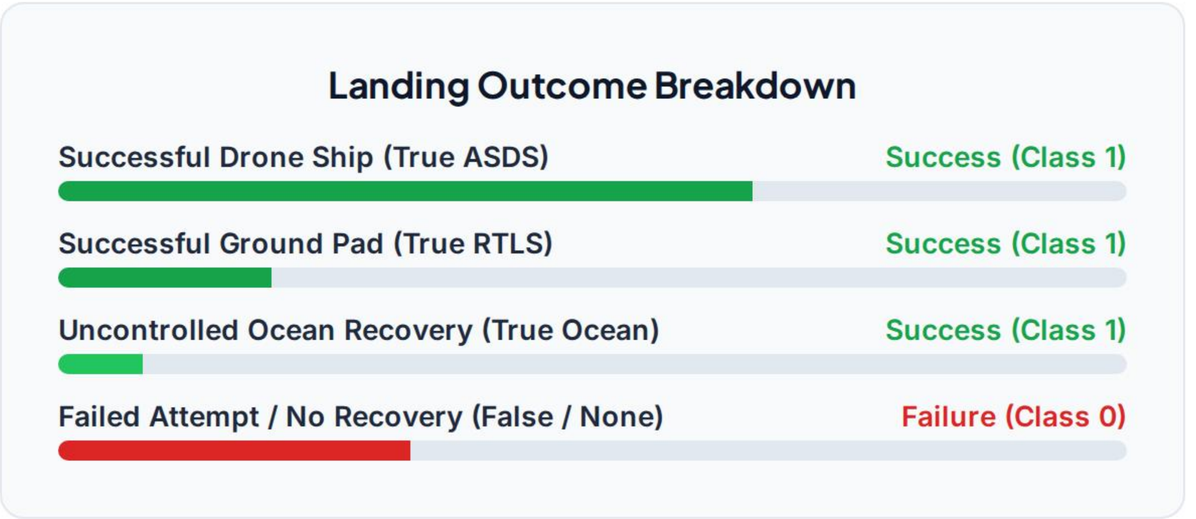
Landing Outcome Classification

Raw launch landing outcomes were categorized into binary training targets: `Class = 1` for successful landings (Ground Pad or Drone Ship), and `Class = 0` for unsuccessful attempts (Ocean crash, unrecovered, or ASDS failure).

> **Target Binary Encoding Strategy:**
> **Class 1 (Success):** `True ASDS`, `True RTLS`, `True Ocean`
> **Class 0 (Failure):** `False ASDS`, `False RTLS`, `False Ocean`, `None ASDS`, `None None`

Missing Value Imputation

Missing `PayloadMass` values were imputed using column average (`df['PayloadMass'].mean()`). One-hot encoding converted categorical features (`Orbit`, `LaunchSite`, `LandingPad`, `Serial`) into 83 numeric features.



EXPLORATORY DATA ANALYSIS

EDA: Flight Number & Payload Relationships

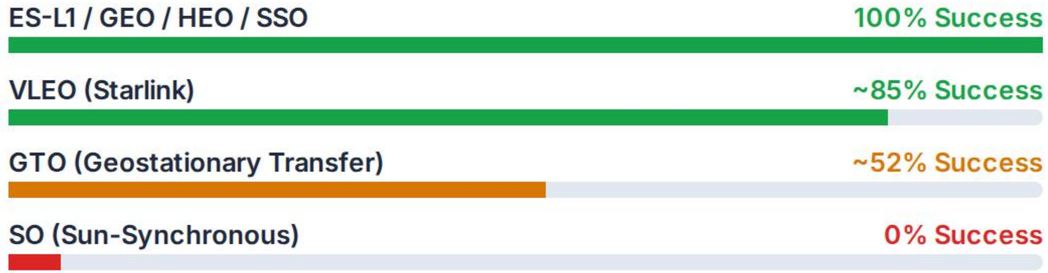
Flight Number vs. Launch Site

Early flights (< 30) were concentrated at CCAFS SLC-40 with lower success rates. As flight number increased (> 30), overall landing success rates drastically improved across all sites.

CCAFS SLC-40	Highest launch volume (Early failures)
KSC LC-39A	**Highest success ratio (~77%)**
VAFB SLC-4E	Polar orbit launches (~67% success)

Orbit Type Success Rates

Success rate varies significantly across orbital destinations. Specialized high-energy orbits demonstrate perfect recovery records.



DATABASE EXPLORATION

EDA with SQL Query Results

Query Objective	SQL Syntax Logic	Analytical Insight / Result
Unique Launch Sites	`SELECT DISTINCT Launch_Site FROM SPACEXTBL`	Identified 4 main sites (CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E)
NASA Payload Sum	`SELECT SUM(PAYLOAD_MASS__KG_) WHERE Customer LIKE 'NASA%'`	Total 45,596 kg carried for NASA resupply missions
Avg Payload (F9 v1.1)	`SELECT AVG(PAYLOAD_MASS__KG_) WHERE Booster_Version LIKE 'F9 v1.1%'`	2,928.4 kg average payload mass for early v1.1 boosters
First Ground Success	`SELECT MIN(Date) WHERE Landing_Outcome = 'Success (ground pad)'`	First successful ground landing achieved on 2015-12-22
Drone Success (4k-6k kg)	`SELECT DISTINCT Booster_Version WHERE Landing_Outcome='Success (drone ship)' AND Payload BETWEEN 4000 AND 6000`	Boosters F9 FT v1.1 B1022, B1026, F9 B4 B1041.1 confirmed successful recovery

Key SQL Takeaway: Landing success rapidly increased post-2016. High maximum payload capacity (15,600 kg) was dominated by Block 5 boosters with 100% successful drone ship landings.

GEOSPATIAL ANALYTICS

Interactive Map Analysis with Folium



Coastline Proximity Safety

All launch sites are situated within **0.8 km to 1.5 km** of the coastline to ensure abort trajectories land over ocean water.

Transport Infrastructure

Sites are positioned adjacent to heavy railways (< 1.2 km) and highways (< 0.5 km) to transport massive booster stages from manufacturing hubs.

City Distance Buffer

Launch pads maintain a **14 km to 23 km** isolation buffer from dense urban centers (e.g., Titusville at 20 km) for public safety.

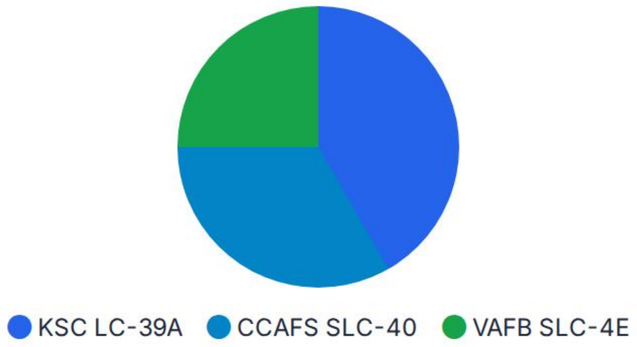
INTERACTIVE DASHBOARD

Plotly Dash Dashboard Visualizations

Site Launch Success Pie Chart

Interactive Dropdown

KSC LC-39A accounts for 41.7% of total successful landing missions, outperforming CCAFS SLC-40 (33.3%) and VAFB SLC-4E (25.0%).



Payload Range vs. Success Correlation

Range Slider (0-10,000kg)

Payloads between **2,000 kg and 5,000 kg** show the highest density of successful landings across all sites. Extremely light (< 2,000 kg) early launches had lower landing success.

0 - 2,000 kg — Moderate Success

2,000 - 5,000 kg — Optimal Success Zone

> 5,000 kg — High Booster Recovery Rate

MACHINE LEARNING METHODOLOGY

Predictive Analysis Setup & Tuning

1

Standard Standardization

Transformed feature matrix `X` using `StandardScaler()` to standardize column distributions for gradient and distance algorithms.

2

Stratified Train/Test Split

Split dataset using `train_test_split(X, Y, test_size=0.2, random_state=2)` to preserve class proportion in train and test sets.

3

GridSearchCV Optimization

Applied 10-fold Cross-Validation over parameter grids to optimize hyperparameters and prevent model overfitting.

4

Confusion Matrix Eval

Evaluated models using classification accuracy, confusion matrices, precision, recall, and ROC-AUC curves on test data.

MODEL PERFORMANCE

Predictive Analysis Classification Results

Algorithm	Best Parameters	Train Accuracy	Test Accuracy
Logistic Regression	`C=0.01, penalty='l2'`	84.6%	83.33%
Support Vector Machine	`kernel='sigmoid', C=1.0`	84.8%	83.33%
Decision Tree	`criterion='gini', max_depth=4`	88.9%	83.33%
k-Nearest Neighbors	`n_neighbors=10, p=1`	84.7%	83.33%

Test Confusion Matrix

3 True Negative (0,0)	3 False Positive (0,1)
0 False Negative (1,0)	12 True Positive (1,1)

STRATEGIC VALUE

Innovative Insights & Reusability Value Engine



Block 5 Reliability Threshold

Booster version iteration (Block 5) achieved near 100% reusability success, reducing statistical failure risk to near zero for post-2018 missions.

Payload Sweet Spot Dynamics

Medium payload weights (2,000–5,000 kg) provide optimal propellant reserves for boostback and entry burns, maximizing drone ship recovery rates.

Commercial Bidding Matrix

Predictive model outputs allow competing space startups to dynamically model SpaceX's profit margin on any given payload configuration.

SUMMARY & RECOMMENDATIONS

Project Conclusions & Future Roadmap

Key Analytical Findings

- **Launch Success Trend:** Reusability rates increased from < 30% in early testing to > 85% in recent operations.
- **Site Efficiency:** KSC LC-39A demonstrates the highest historical landing recovery rate.
- **Model Accuracy:** ML classifiers reliably predict landing outcome with 83.33% test set accuracy.

Actionable Recommendations

- **Target Payload Range:** Optimize commercial client bids for GTO/LEO payloads within the 2,000–5,000 kg window.
- **Launch Pad Strategy:** Prioritize missions operating out of KSC LC-39A for maximum flight reliability.
- **Real-Time Telemetry Expansion:** Incorporate real-time weather and wind vector features to enhance model sensitivity.

Peer Review Checklist Complete
All notebooks, dashboards, SQL queries, maps, and models validated and submitted.

40 / 40 Points Target

Thank You!

Data Science Capstone Project complete. Questions & discussions welcomed.

GITHUB REPOSITORY

github.com/spacex-capstone-report

PDF PRESENTATION EXPORT

SpaceX_Capstone_Final_Report.pdf

Image Sources



https://cdn.geekwire.com/wp-content/uploads/2020/05/20200530_SpaceX_Launch_807-2-1.jpg

Source: www.geekwire.com



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